

AI ENGINEER CAREER ROADMAP

Student → AI Engineer | Timeline: 3 Months | Starting Skill: Python

Generated via Chain-of-Thought Reasoning

CURRENT POSITION

Student

Skill: Python | Level: Beginner–Intermediate

TARGET GOAL

AI Engineer

Timeline: 3 Months | Focus: LLMs, APIs, Deployment

SKILL GAP ANALYSIS

You Have

- Python basics
- Problem-solving mindset
- Learning agility

Critical Gaps

- ML / Deep Learning
- LLM & Prompt Engineering
- API Dev (FastAPI)

Nice to Have

- MLOps / Docker
- Vector DBs
- Cloud (AWS/GCP)

LEARNING PLAN

Wk 1-2	Python for ML (NumPy, Pandas, Sklearn)
Wk 3-4	Deep Learning basics (PyTorch / Keras)
Wk 5-6	LLMs: OpenAI API, LangChain, RAG
Wk 7-8	FastAPI + REST API for AI models
Wk 9-10	Vector DBs (Pinecone/Chroma) + RAG apps
Wk 11-12	Deploy on HuggingFace / Render / AWS

NETWORKING STRATEGY

LinkedIn: Post weekly AI project builds + learnings
GitHub: Push all projects; maintain green streaks
Twitter/X: Follow: Andrej Karpathy, Harrison Chase
Discord: Join: LangChain, Hugging Face communities

SUGGESTED PROJECTS

- AI Chatbot with OpenAI API + memory
- PDF Q&A RAG App (LangChain + Chroma)
- Image Caption Generator (CLIP model)
- Voice-to-Summary App (Whisper API)
- ML Dashboard deployed on Streamlit
- FastAPI AI microservice on Render

IMMEDIATE NEXT ACTIONS

- Set up GitHub profile TODAY
- Enroll: fast.ai or DeepLearning.ai (free)
- Build Chatbot in Week 2 — share on LinkedIn
- Apply to AI internships from Month 2

MONTHLY MILESTONES

Month 1

LEARN

- Complete ML + DL fundamentals
- Build 2 mini-projects
- Setup GitHub + LinkedIn

Month 2

BUILD

- Build RAG App + FastAPI service
- Deploy 1 live AI project
- Start internship applications

Month 3

LAUNCH

- Portfolio of 4–6 AI projects
- Apply to 20+ AI Engineer roles
- Target 2 interviews/week